



KBCC

Boron Carbide Composite

KEY PROPERTIES

- Extremely high hardness
- Low density
- Suitable for EDM machining
- High wear resistance

APPLICATIONS

- Cutting tools
- Watch-making industry
- Wear resistant parts

MAIN PROPERTIES

Property	Value	Unit
Density	3.0	g/cm ³
Flexural strength	635	MPa
Hardness	32	GPa
Fracture Toughness	3.5	MPa·√m

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air) | 800 °C

Recomended temp | <750 °C

Continuous temp | 700 °C

ELECTRICAL BEHAVIOR

Semi-conductive

MATERIAL INFORMATION

Material type | Boron Carbide composite

Structure | Dense ceramic

Porosity level | EDM machinable

Composition | Ultra-high hardness

Color | Black

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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